Lecture 8

→ Solving $Ax = b$

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \quad \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Augmented matrix →

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_1 - b_2 + 2b_1 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & \underbrace{b_3 - b_2 - b_1}_{\to 0} \end{array}\right]$$

$$\boxed{b_3 = b_1 + b_2}$$ condition for solvability.

\# $Ax = b$ is solvable when b is in the column space of A. $C(A)$.

$\Updownarrow$

→ If there is a combination of rows that gives 0, that component of b also has to be 0.

• Finding the solution ->

① A particular solⁿ: Set all free vars to 0

$$x_{particular} = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} \quad \text{for} \quad b = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$$

② Add the nullspace to $x_{particular}$ will give us the complete solⁿ.

$$X = X_p + N(A)$$

• For the given example $\begin{bmatrix} -2 & 2 \\ 1 & 0 \\ 0 & -2 \\ 0 & 1 \end{bmatrix}$

$$N(A) = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

→ This set of solutions form a 2-dimensional solution space which is not a subspace

General picture → $m \times n$ of rank r

$$\boxed{r \leq m \quad \&\& \quad r \leq n}$$

because each row can only have one pivot, so $\underline{r \leq \min(m, n)}$

- <u>Full</u> <u>column</u> <u>rank</u> $[r = n]$
    - Every column has a pivot ✓
    - No free variables. ✓
    - Origin is the only null space. ✓
    - <u>One</u> or no solutions? ✓

eg

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{bmatrix} \qquad \underline{\underline{\text{rank}(A) = 2}}$$

$$\text{rref} \Rightarrow \left[\begin{array}{cc|c} 1 & 1 & b_1 \\ 0 & -1 & b_2 - 2b_1 \\ 0 & -2 & b_3 - 3b_1 \\ 0 & -3 & b_4 - 4b_1 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & b_1 + b_2 - 2b_1 \\ 0 & -1 & b_2 - 2b_1 \\ 0 & 0 & b_3 - 3b_1 - 2b_2 + 4b_1 \\ 0 & 0 & b_4 - 4b_1 - 3b_2 + 6b_1 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 0 & b_2 - b_1 \\ 0 & 1 & -b_2 + 2b_1 \\ 0 & 0 & b_3 - 2b_2 + b_1 \\ 0 & 0 & b_4 - 3b_2 + 2b_1 \end{array}\right]$$

- Full row rank $[r = m]$

  - Every row has a pivot ✓
  - No zero rows $\Rightarrow N(A) = \{0\}$
  - Free vars would exist in rectangular matrix. $n - r \rightarrow$ free vars
  - Either 1 or many solutions.
    ↳ there always exists a solution

eg
$$\begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix}$$

$$\text{ref} = \begin{bmatrix} 1 & 2 & 6 & 5 \\ 0 & -5 & -17 & -14 \end{bmatrix}$$

$$\text{rref} = \begin{bmatrix} 1 & 0 & 6 - 34/5 & 5 - 28/5 \\ 0 & 1 & 17/5 & 14/5 \end{bmatrix}$$

$$\text{rref} = \begin{bmatrix} 1 & 0 & -4/5 & -3/5 \\ 0 & 1 & 17/5 & 14/5 \end{bmatrix}$$

→ Square matrix $[r = m = n]$
Full rank

- $N(A) =$ origin
- Exactly one sol^n exists for every **b**
- No free vars.
- invertible !!

eg
$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \xrightarrow[\text{After some steps}]{} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow I$$

Imp

- The rank of a matrix tells us everything about the number of solutions.